
WEB SISTEMAK

2025/2026 IKASTURTEA

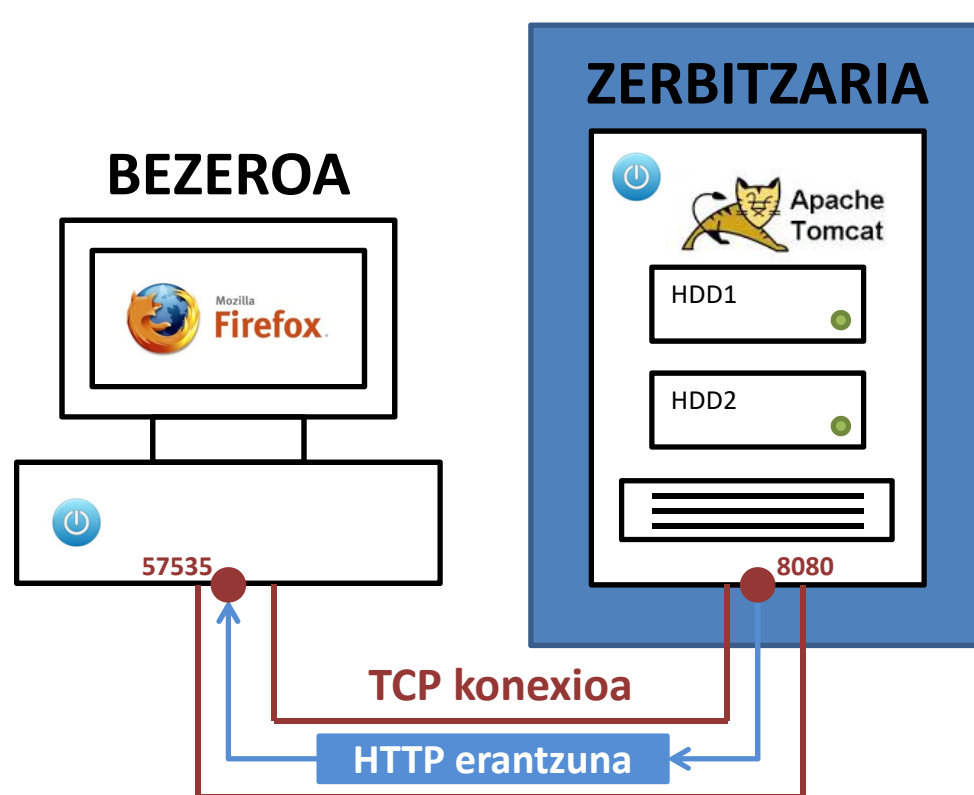
Tomcat web zerbitzaria

Sarrera eta adibideak
Bista eta logikaren arteko banaketa
Datu bilketa inprimaki formatuan



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WEB ZERBITZARIA: HTTP ESKAERAK PROZESATZEN DITU ETA HTTP ERANTZUNAK ITZULTZEN DITU



HTTP erantzunaren sintaxia

HTTP/1.1 Status Deskribapena
Goiburuak
CRLF
Mezuaren gorputza (zortzikoteetan)

HTTP erantzun baten adibidea

```
HTTP/1.1 200 OK
Date: Thu, 20 Nov 2015 20:25:52 GMT
Last-Modified: Tue, 17 Sep 2015 13:00:02 GMT
ETag: "1a968-3ec-4e693e61bb8b6"
Content-Length: 76
Content-Type: text/html; charset=ISO-8859-1

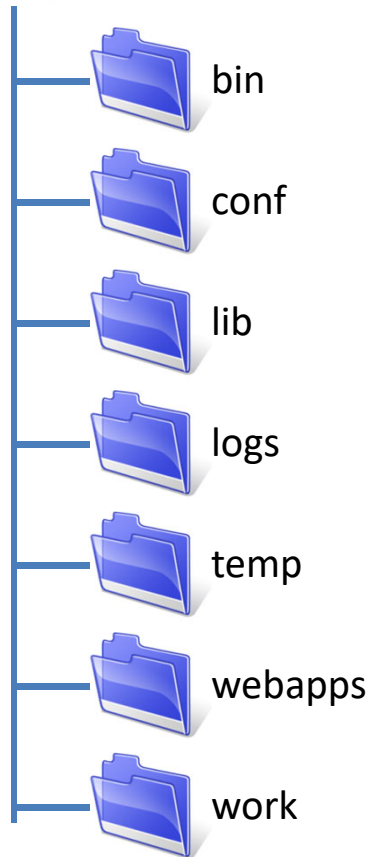
<html><head><title>index.html</title></head>
<body>Hello World!</body></html>
```

WEB ZERBITZARIA: TOMCAT

→ DIREKTORIO EGITURA



CATALINA_HOME



Tomcat erabiltzeko script-ak gordetzen ditu, adibidez: Tomcat abiarazi eta gelditzeko script-ak.

Tomcat-en konfigurazio orokorra adierazten duten ezaugarri-fitxategiak gordetzen ditu.

Tomcat-en bihotza osotzen duten liburutegiak gordetzen ditu, adibidez: servlet API-a. Liburutegi hauek web aplikazio guztientzako eskuragarri daude.

Tomcat-en arazketa fitxategiak (log-ak) gordetzen ditu.

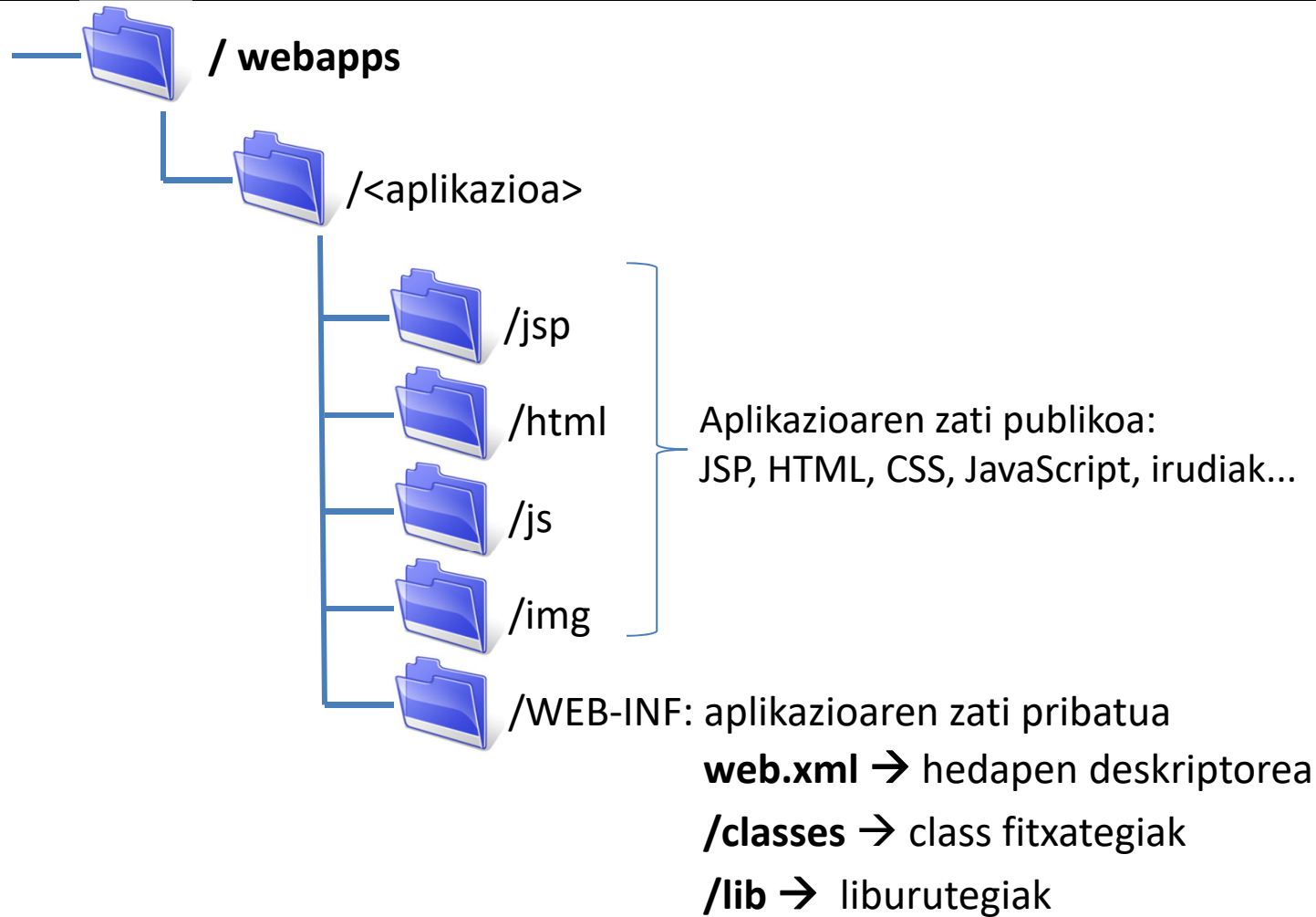
Tomcat-en aldi baterako fitxategiak gordetzen ditu.

Hemen web aplikazioak hedatzen dira.

Tomcat-ek tarteko ekintzak burutzean sortzen dituen fitxategiak gordetzen ditu, adibidez: JSP-ak konpilatzerakoan sortzen den servlet-ak.

WEB ZERBITZARIA: TOMCAT

→ WEBAPP BATEN EGITURA

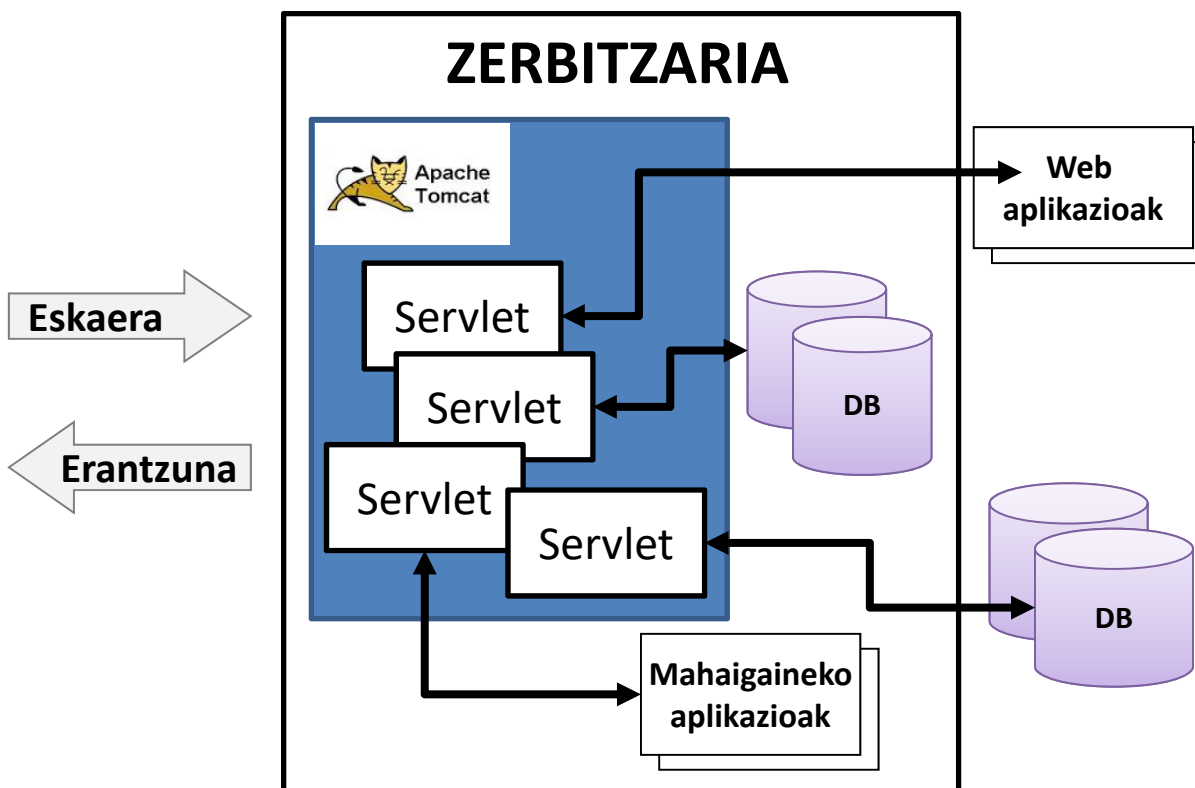
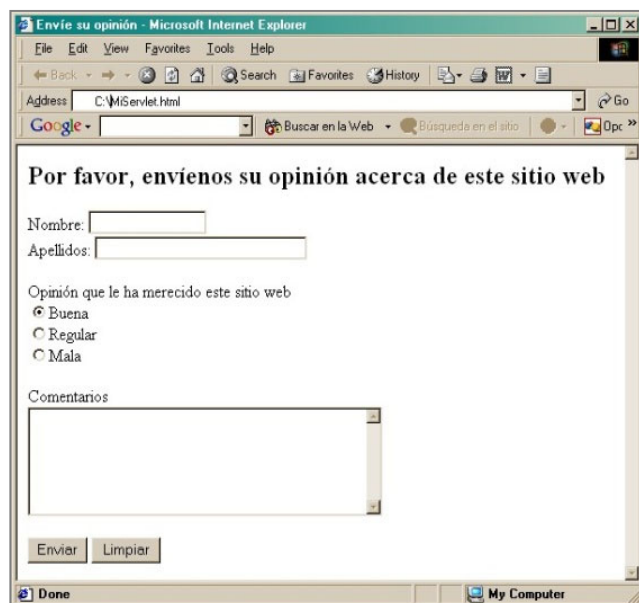


TOMCAT: WEB APLIKAZIO BATEN HEDAPENA

- **Zuzenean:** web aplikazioaren direktorio egitura sortu eta fitxategi bakoitza dagokion direktorioan sartu.
- **WAR fitxategia**
 - Web aplikazioaren direktorioa ZIP formatuan konprimatu eta paketatzerakoan eratortzen den fitxategia da.
 - IntelliJ-k WAR fitxategiak automatikoki sortzen ditu:
 - Proiektua → target → WAR fitxategia

SERVLET-AK

- Servlet-a web bezero baten eskaera prozesatu eta erantzuten dituen programa bat da.
- Web aplikazio bat hainbat servlet-ez osaturik egon daiteke.
- Tomcat servlet edukitzailea da.



SERVLET-AK

- Servlet-ak mota askotako eskaerei erantzun diezaieke, baina orokorrean web zerbitzariak jasotzen dituzten HTTP eskaerak prozesatu eta erantzuteko erabiltzen dira.
- Eginkizun honetarako Java Servlet teknologiak API jakin bat darabil: **javax.servlet** eta **javax.servlet.http** interfazeetan definiturikoa. Tomcat API honen inplementazio bat da.

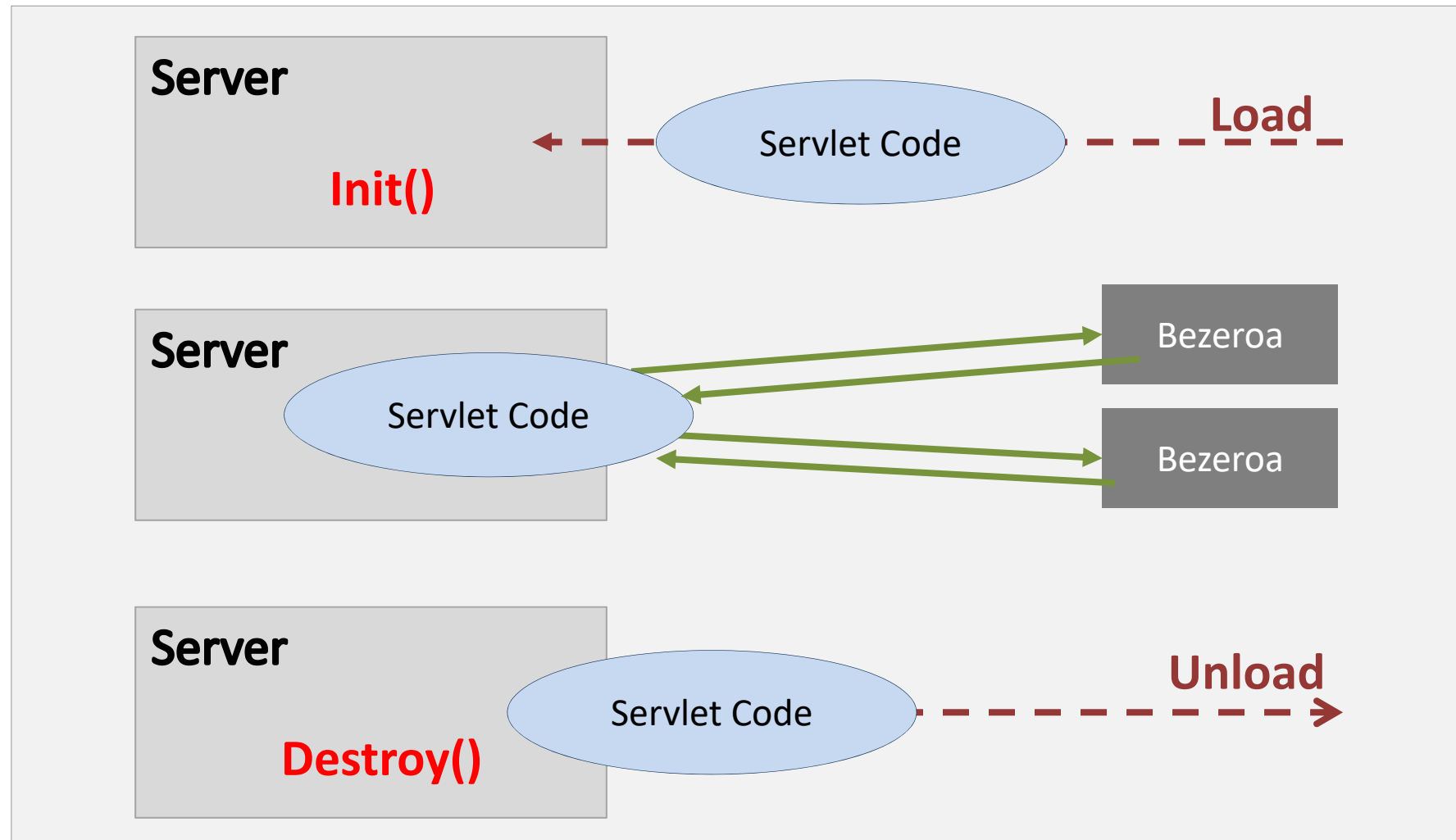
– *javax.servlet*

- <http://docs.oracle.com/javaee/5/api/javax/servlet/package-summary.html>

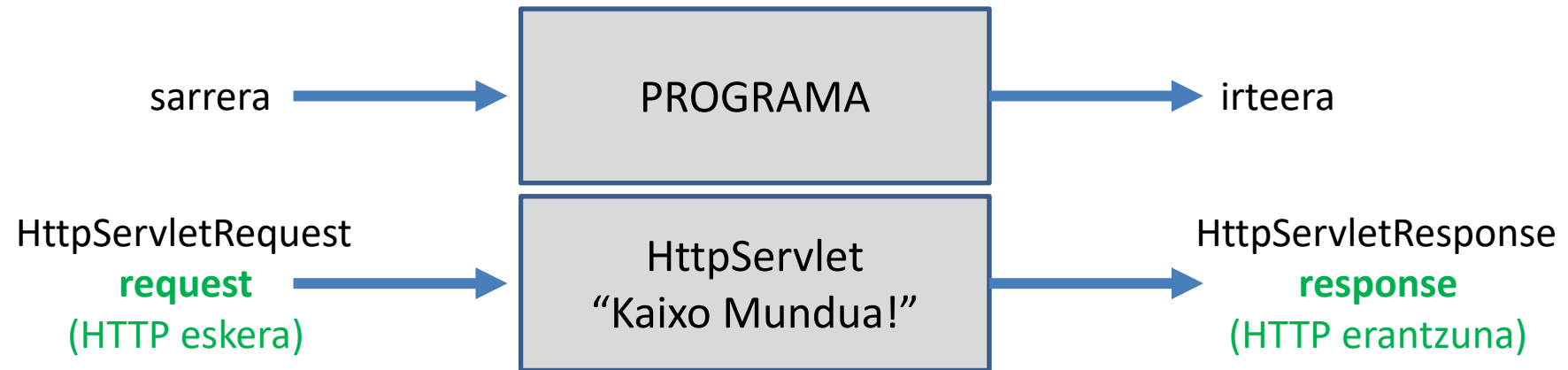
– *javax.servlet.http*

- <http://docs.oracle.com/javaee/5/api/javax/servlet/http/package-summary.html>

SERVLET-AK: BIZITZA ZIKLOA



SERVLET-AK: NIRE LEHENENGO SERVLET-A



```
package nirepaketea;
```

```
import java.io.*;
```

```
import javax.servlet.*;
```

```
import javax.servlet.http.*;
```

Servlet API-a

```
public class KaixoMundua extends HttpServlet {
```

HTTP motako servlet-a

```
    public void doGet(HttpServletRequest request, HttpServletResponse response)
```

```
        throws ServletException, IOException {
```

```
        PrintWriter http_out = response.getWriter();
```

```
        http_out.println("Kaixo Mundua!");
```

```
    }
```

```
}
```

SERVLET-AK: NIRE LEHENENGO SERVLET-A

→ web.xml

```
<?xml version="1.0" encoding="UTF-8"?>

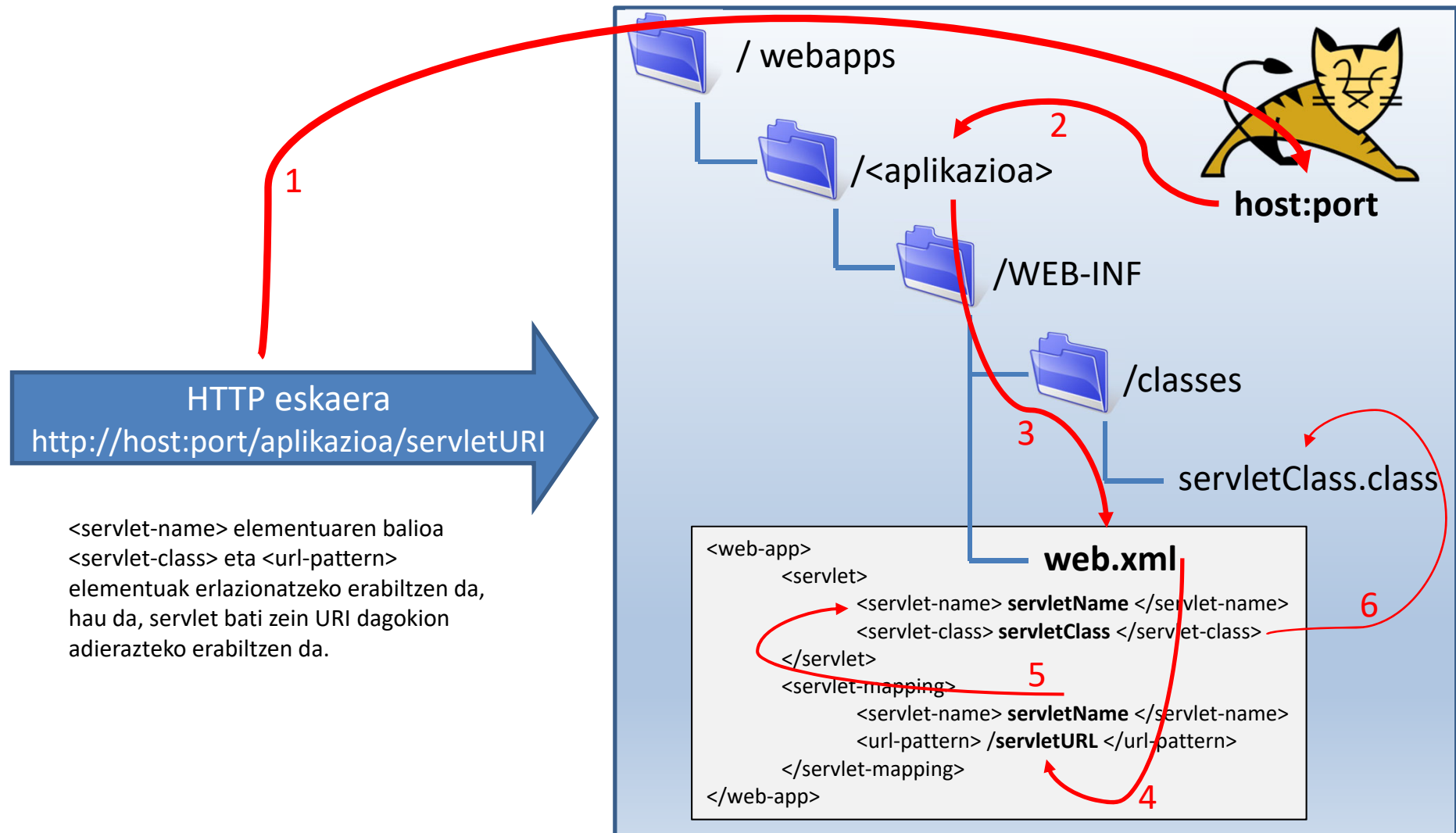
<web-app xmlns:xsi=http://www.w3.org/2001/XMLSchema-instance
  xmlns=http://xmlns.jcp.org/xml/ns/javaee
  xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
    http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd" id="WebApp_ID" version="3.1">

  <servlet>
    <servlet-name>KaixoMundua</servlet-name>
    <servlet-class>nirepaketea.KaixoMundua</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>KaixoMundua</servlet-name>
    <url-pattern>/servlet/kaixomundua</url-pattern>
  </servlet-mapping>

</web-app>
```

SERVLET-AK: NIRE LEHENENGO SERVLET-A

→ web.xml



SERVLET-AK: NIRE LEHENENGO SERVLET-A

→ ARAZKETA

```
package nirepaketea;

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class KaixoMundua extends HttpServlet {

    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        System.out.println("---> Entering KaixoMundua servlet"); // log

        PrintWriter http_out = response.getWriter();
        http_out.println("Kaixo Mundua!");

        System.out.println("<--- Exiting KaixoMundua servlet"); // log
    }
}
```

SERVLET-AK: NIRE LEHENENGO SERVLET-A

→ ARAZKETA

The screenshot shows the Eclipse IDE interface. The top toolbar includes icons for File, Edit, Navigate, Search, Project, Run, Window, and Help. The Project Explorer on the left shows the project structure for 'KaixoMundua' and 'proba'. The main editor displays the 'main_db.java' file. The browser window shows the URL 'http://localhost:8080/KaixoMundua/servlet/kaixomundua' and the output 'Kaixo Mundua!'. The Console window at the bottom shows the Tomcat v8.5 Server logs, including the startup sequence and the execution of the 'KaixoMundua' servlet. The logs are as follows:

```
Tomcat v8.5 Server at localhost [Apache Tomcat] C:\Program Files\Java\jdk1.8.0_77\bin\javaw.exe (Feb 27, 2017, 10:21:36 PM)
INFO: Starting service Catalina
Feb 27, 2017 10:21:39 PM org.apache.catalina.core.StandardEngine startInternal
INFO: Starting Servlet Engine: Apache Tomcat/8.5.11
Feb 27, 2017 10:21:40 PM org.apache.jasper.servlet.TldScanner scanJars
INFO: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JARs t
Feb 27, 2017 10:21:40 PM org.apache.catalina.util.SessionIdGeneratorBase createSecureRandom
INFO: Creation of SecureRandom instance for session ID generation using [SHA1PRNG] took [184] milliseconds.
Feb 27, 2017 10:21:41 PM org.apache.jasper.servlet.TldScanner scanJars
INFO: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JARs t
Feb 27, 2017 10:21:41 PM org.apache.jasper.servlet.TldScanner scanJars
INFO: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JARs t
Feb 27, 2017 10:21:41 PM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler [http-nio-8080]
Feb 27, 2017 10:21:41 PM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler [ajp-nio-8009]
Feb 27, 2017 10:21:41 PM org.apache.catalina.startup.Catalina start
INFO: Server startup in 2817 ms
----> Entering KaixoMundua servlet
----> Exiting KaixoMundua servlet
```

SERVLET-AK: HTTPSERVLETREQUEST: METODOAK

<http://docs.oracle.com/javaee/6/api/javax/servlet/http/HttpServletRequest.html>

- String **getMethod()**
- String **getHeader**(String name)
- java.util Enumeration<String> **getHeaderNames()**
- String **getParameter**(String name)
- java.util Enumeration<String> **getParameterNames()**
- ...

SERVLET-AK: HTTPSERVLET**RESPONSE**: METODOAK

<http://docs.oracle.com/javaee/6/api/javax/servlet/http/HttpServletResponse.html>

- void **setStatus**(int sc)
- void **setContentType**(String type)
- void **setContentLength**(String type)
- void **addHeader**(String name, String value)
- void **addCookie**(Cookie cookie)
- java.io.PrintWriter **getWriter**()
- void **sendRedirect**(String location) → 302 status kodea bueltatzen du
- ...

SERVLET-AK: NIRE BIGARREN SERVLET-A

→ HTTP ESKAERAREN GOIBURUAK IRAKURRI

The screenshot shows the Eclipse IDE interface. The Project Explorer on the left displays the project structure for 'KaixoMundua', including 'src' (with 'nirepaketea' containing 'KaixoMundua.java' and 'KaixoMunduaGoiburuak.java'), 'Libraries', 'JavaScript Resources', 'build', 'WebContent', 'META-INF', 'WEB-INF' (with 'lib' and 'web.xml'), and 'proba' (with 'main_db.java' and 'main.java'). The main editor shows the 'KaixoMunduaGoiburuak.java' file. The Console window at the bottom displays the following output:

```
Tomcat v8.5 Server at localhost [Apache Tomcat] C:\Program Files\Java\jdk1.8.0_77\bin\javaw.exe (Feb 27, 2017, 10:18:24 PM)
INFO: Starting ProtocolHandler [http-nio-8080]
Feb 27, 2017 10:18:30 PM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler [ajp-nio-8009]
Feb 27, 2017 10:18:30 PM org.apache.catalina.startup.Catalina start
INFO: Server startup in 2848 ms
----> Entering KaixoMunduaGoiburuak servlet
    HTTP eskaeraren goiburuak:
        accept: image/gif, image/jpeg, image/pjpeg, application/x-ms-application, application/xhtml+xml, application/x-ms-xbap, */*
        accept-language: en-US,en;q=0.8,es-ES;q=0.5,es;q=0.3
        cache-control: no-cache
        ua-cpu: AMD64
        accept-encoding: gzip, deflate
        user-agent: Mozilla/5.0 (Windows NT 6.2; Win64; x64; Trident/7.0; rv:11.0) like Gecko
        host: localhost:8080
        connection: Keep-Alive
<--- Exiting KaixoMunduaGoiburuak servlet
```


SERVLET-AK: NIRE HIRUGARREN SERVLET-A

→ HTML IRTEERA SERVLET-AREN BARNEAN

```
package nirepaketea;

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class KaixoMunduaHTMLhardcoded extends HttpServlet{

    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        System.out.println("---> Entering KaixoMunduaHTMLhardcoded servlet");

        response.setContentType("text/html");

        PrintWriter http_out = response.getWriter();
        http_out.println("<html>");
        http_out.println("<head><title>HTML servlet</title></head>");
        http_out.println("<body>");
        http_out.println("<h2>HTML irteera sortzen duen servlet-a</h2>");
        http_out.println("<p>Servlet honek bere barnean kodifikatuta duen");
        http_out.println(" HTML dokumentua itzultzen du.</p>");
        http_out.println("</body>");
        http_out.println("</html>");

        http_out.flush();
        http_out.close();

        System.out.println("<--- Exiting KaixoMunduaHTMLhardcoded servlet");
    }
}
```

SERVLET-AK: NIRE HIRUGARREN SERVLET-A

→ HTML IRTEERA SERVLET-AREN BARNEAN

- HTML eta Java kodea nahasteak dakarren desabantaila: bista eta kontrola nahasten dira.
 - HTML-an egindako aldaketa bat berehala ikusi beharko litzakete, baina servletaren barnean egoterakoan:
 - HTML-ean sartutako aldaketak bistaratzeko servleta konpilatu behar da.
 - Zerbitzariaren berrabiarazi beharra azaldu daiteke, berrabiarazteak irauten duen denboran erabiltzaileak saioa galtzea eta zerbitzu gabe uztea suposatzen duelarik.

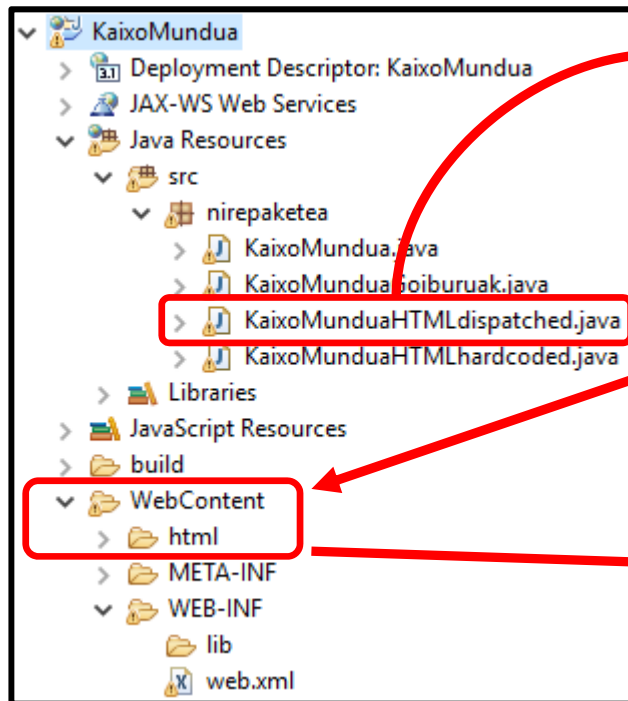
SERVLET-AK: NIRE LAUGARREN SERVLET-A

→ BISTA ETA KONTROLA BANANDU

Nabigatzaileak *WebContent* karpeta jartzen den edukia zuzenean atzitu dezake (*WEB-INF* izan ezik)

HTML orriak *WebContent* karpeta barnean, *html* izeneko karpeta batean, jartzea gomendatzen da.

Horrela, HTML-an egindako aldaketak nabitzailen zuzenean isladatzen dira orria birkargatzerakoan.



```
package nirepaketea;

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class KaixoMunduaHTMLdispatched extends HttpServlet{

    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        System.out.println("---> Entering KaixoMunduaHTMLdispatched servlet");

        response.setContentType("text/html");

        System.out.println("\tRedirecting to HTML document");
        RequestDispatcher rd = request.getRequestDispatcher("/html/weborria.html");
        rd.forward(request, response);

        System.out.println("<--- Exiting KaixoMunduaHTMLdispatched servlet");
    }
}
```

```
<html>
  <head>
    <TITLE>HTML servlet</TITLE>
  </head>
  <body>
    <h2>HTML irteera sortzen duen servlet-a</h2>
    <p>Servlet honek irteera aparteko HTML batera birbideratzen du.</p>
  </body>
</html>
```

SERVLET-AK: NIRE BOSTGARREN SERVLET-A

→ INPRIMAKI FORMATUDUN DATU BILKETA

```
<!DOCTYPE html>
<html>
<head>
  <title>DoGet o DoPost</title>
</head>
<body>
  <form method="GET" action="../servlet/holamundoGetPost">
    Nombre: <input type="text" name="nombre" >
    <button type="submit">DoGet</button>
  </form> <br>
  <form method="POST" action="../servlet/holamundoGetPost">
    Nombre: <input type="text" name="nombre"/>
    <button type="submit">DoPost</button>
  </form>
</body>
</html>
```

```
package micarpeta;

import java.io.*; import javax.servlet.*;import javax.servlet.http.*;

public class HolaMundoGetPost extends HttpServlet {

  public void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    System.out.println("---> Entering HolaMundoGetPost servlet GET");

    String nombre = request.getParameter("nombre");
    PrintWriter out = response.getWriter();
    try {
      out.println("Hola, " + nombre);
      out.println("Has pasado tu nombre con GET en la URI.");
    } finally {
      out.close();
    }

    System.out.println("---- Exiting HolaMundoGetPost servlet GET");
  }

  public void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    System.out.println("---> Entering HolaMundoGetPost servlet POST");

    String nombre = request.getParameter("nombre");
    PrintWriter out = response.getWriter();
    try {
      out.println("Hola, " + nombre);
      out.println("Has pasado tu nombre con POST en el cuerpo de la petición.");
    } finally {
      out.close();
    }

    System.out.println("---- Exiting HolaMundoGetPost servlet POST");
  }
}
```

SERVLET-AK: NIRE BOSTGARREN SERVLET-A

→ NAN LETRA KALKULATU

```
<servlet>
  <servlet-name>nanLetra</servlet-name>
  <servlet-class>nirepaketea.nanLetra</servlet-class>
</servlet>

<servlet-mapping>
  <servlet-name>nanLetra</servlet-name>
  <url-pattern>/servlet/nanLetra</url-pattern>
</servlet-mapping>
```

```
<!DOCTYPE html>
<html>
<head>
  <title>DoGet o DoPost</title>
</head>
<body>
  <form method="POST" action="/proba/servlet/nanLetra">
    Zure NAN zenbakia: <input type="text" name="nan"/>
    <input type="submit" value="bidali"/>
  </form></body>
</html>
```

```
package nirepaketea;

import java.io.*; import javax.servlet.*; import javax.servlet.http.*;

public class nanLetra extends HttpServlet {

  protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    int nanZenbakia = Integer.parseInt(request.getParameter("nan"));
    String karakterak = "TRWAGMYFPDXBNJZSQVHLCKE";
    int hondarra = nanZenbakia % 23;
    char nanLetra = karakterak.charAt(hondarra);

    PrintWriter http_out = response.getWriter();
    http_out.println(nanLetra);
  }

  protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    doGet(request, response);
  }
}
```